



Glymphatic function in humans: indicators, modulators, and measurement strategies

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Abstract

Purpose The glymphatic system is an important pathway for brain waste clearance and is increasingly implicated in neurodegenerative, cerebrovascular, traumatic, and sleep-related disorders. Clinical translation remains challenging because glymphatic function cannot currently be comprehensively or directly quantified in vivo in humans, and assessment relies on heterogeneous measures differing in physiological specificity and measurement directness. This review integrates human glymphatic indicators, physiological modulators, and measurement strategies within a unified framework.

Methods A narrative literature review was conducted using human studies and key foundational research identified through PubMed and Google Scholar. Assessment methods were organized according to the authors' proposed three-domain framework of glymphatic physiology (transport, exchange, and clearance outcome), recognizing that modalities may span multiple domains. Major physiological modulators and confounders were summarized, and measurement strategies were evaluated based on feasibility, reproducibility, and translational applicability.

Results Human glymphatic assessment relies on complementary approaches with different levels of measurement directness, including tracer-informed kinetic MRI; neurofluid proxies such as BOLD–CSF coupling and CSF flow MRI; indirect imaging correlates such as DTI-ALPS and enlarged perivascular spaces; and downstream biological correlates such as CSF solute biomarkers. Measurements can be influenced by sleep, circadian timing, respiration, cardiovascular pulsatility, posture, medications, vascular risk factors, and aging. A domain-first, tiered framework is proposed to support multimodal evaluation and interpretation.

Conclusion The proposed framework organizes multimodal glymphatic-related measures according to physiological domain and measurement directness. Standardized acquisition, harmonized analytical methods, rigorous confounder reporting, and prospective validation may improve reproducibility and support future clinical translation.

Keywords Glymphatic system · DTI-ALPS · BOLD–CSF coupling · Neuroimaging biomarkers · Cerebrospinal fluid · Measurement strategies

Introduction

Glymphatic system: overview and clinical relevance

The glymphatic system has emerged as a central concept in understanding brain waste clearance and neurofluid homeostasis. The glymphatic system (Fig. 1) is a brain-wide

clearance pathway in which cerebrospinal fluid (CSF) enters the brain through perivascular spaces (PVS), exchanges with interstitial fluid (ISF), and facilitates the removal of metabolic byproducts from the central nervous system. This process is closely linked to astrocytic aquaporin-4 (AQP4), vascular dynamics, and brain state [1–3]. However, important aspects of glymphatic physiology remain under investigation, including the mechanisms of fluid and solute transport and their translation to humans [1, 3].

Evidence increasingly suggests that glymphatic dysfunction contributes to the pathophysiology of neurological disorders through impaired waste clearance, neuroinflammation, and neurovascular dysfunction [4]. These include neurodegenerative diseases such as Alzheimer's disease

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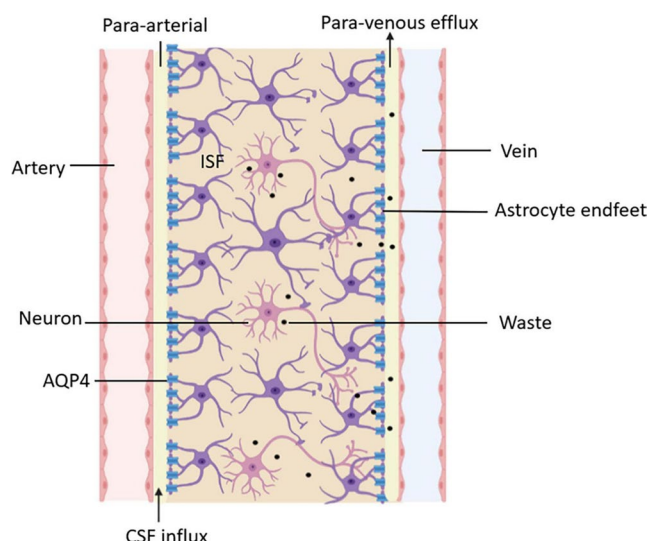


Fig. 1 Schematic illustration of the canonical glymphatic model. Cerebrospinal fluid (CSF) is proposed to enter the brain along periarterial perivascular spaces, interact with interstitial fluid (ISF) within the parenchyma at an interface involving astrocytic endfeet enriched in aquaporin-4 (AQP4), and contribute to downstream fluid and solute drainage through perivenous and extracranial pathways. Reprinted from Ding Z, Fan X, Zhang Y, Yao M, Wang G, Dong Y, Liu J, Song W. The glymphatic system: a new perspective on brain diseases. *Front Aging Neurosci.* 2023;15:1179988. <https://doi.org/10.3389/fnagi.2023.1179988>. Distributed under the terms of the Creative Commons Attribution (CC BY) License [1]

and Parkinson's disease, cerebrovascular disorders including stroke and cerebral small vessel disease, traumatic brain injury, and sleep-related disorders [3–5]. Because glymphatic-related neurofluid physiology is strongly state-dependent, particularly across sleep and wakefulness, considerable research interest has focused on the potential use of glymphatic-related biomarkers for disease assessment and mechanistic investigation [3, 4, 6].

Challenges in human assessment and sources of variability

Despite its potential relevance, glymphatic function cannot currently be comprehensively or directly quantified in vivo in humans, and human studies cannot reproduce the same tracer-based approaches that have established much of the mechanistic evidence in animal models [7]. Human research therefore relies on heterogeneous measures that differ in their degree of directness, including neuroimaging-derived measures, physiological signals, and downstream biological markers [7, 8]. These approaches can be distinguished according to an evidence hierarchy comprising (1) direct or tracer-informed measures of CSF distribution and clearance, (2) physiology-sensitive neurofluid proxies, (3) indirect imaging correlates, and (4) downstream biological correlates. These categories provide different levels of

evidence regarding glymphatic-related physiology and should not be interpreted as interchangeable measures of glymphatic function. Importantly, these approaches capture only partial components or correlates of the pathway, such as perivascular diffusion patterns, CSF flow dynamics, structural remodeling of PVS, or clearance-related outcomes [7]. This limitation makes it difficult to interpret any single measure as a complete or specific representation of glymphatic function.

Inconsistencies in the literature are common and often expected. Glymphatic-related signals vary with sleep stage and arousal, circadian timing, respiration and CO₂ fluctuations, blood pressure and cardiovascular pulsatility, anesthetic exposure, and aging [3, 9, 10]. In addition, different tools interrogate different physiological domains and provide different levels of measurement directness, so diffusion-based markers, functional coupling metrics, tracer kinetic approaches, structural imaging indicators, and downstream biological measures are not interchangeable. Variation in acquisition protocols, analysis pipelines, and confounder reporting further reduces reproducibility across studies.

Rationale, aim, and literature approach

Existing reviews have frequently focused on imaging markers, physiological mechanisms, or disease-specific associations in isolation [4, 5, 8]. However, there remains no unified framework for human glymphatic assessment integrating what is measured (indicators), what influences measurements (modulators and confounders), and how assessment is performed in practice (measurement strategies). This fragmentation limits comparability across modalities and slows clinical translation.

This narrative review synthesizes current human glymphatic-related indicators and proxies, major modulators and confounders that influence human measurements, and practical measurement strategies for research and translational application. We also propose a tiered assessment framework with minimum reporting recommendations to support study design, interpretation, and translational application. This framework represents the authors' synthesis of the reviewed literature rather than a formal consensus guideline.

A narrative literature search was conducted in PubMed and Google Scholar, with the final search performed on September 3, 2026, covering publications available through August 2026. Representative search combinations included 'glymphatic' with 'human,' 'MRI,' 'CSF,' 'PVS,' 'DTI-ALPS,' 'BOLD-CSF,' 'tracer MRI,' 'sleep,' 'respiration,' 'posture,' 'aging,' 'reproducibility,' and 'biomarker.' Recent human studies were prioritized for assessment methods, physiological associations, feasibility, and reproducibility.

Earlier animal and human studies were included when they provided foundational mechanistic evidence or introduced major assessment methods. Foundational studies were identified by citation tracing from relevant reviews and primary studies, with preference given to original reports introducing a method or establishing a key physiological finding. Because this was a narrative review, formal systematic inclusion and exclusion criteria were not prespecified. Studies were selected based on their relevance to human glymphatic-related assessment, physiological modulators, methodological validity, or foundational mechanistic evidence, with preference given to peer-reviewed original studies when method-specific or mechanistic claims were made. Studies informing the proposed framework and reporting recommendations were selected based on recurring methodological and reproducibility considerations. Accordingly, the search was intended to identify representative and methodologically important literature rather than provide exhaustive coverage.

Defining glymphatic function for human assessment

Core pathway and compartments

The glymphatic system describes a perivascular fluid transport network that supports brain clearance through coordinated movement of CSF and ISF [1, 3, 7]. Foundational *in vivo* tracer studies by Iliff et al. demonstrated in mice that subarachnoid CSF enters the brain along periaxonal paravascular spaces, exchanges with interstitial fluid within the parenchyma, and contributes to interstitial solute clearance along paravenous pathways [11]. In the canonical glymphatic model, CSF is proposed to enter the brain along periaxonal PVS, interact with ISF within the parenchyma, and subsequently contribute to solute drainage through perivenous and extracranial pathways, including meningeal lymphatic vessels [12, 13]. However, important aspects of this model remain under investigation, including the relative contributions of convective and diffusive transport, the organization and directionality of periaxonal and perivenous pathways, and the extent to which mechanisms demonstrated in animal tracer studies translate directly to humans [3, 7]. Brain clearance may also involve overlapping perivascular, meningeal lymphatic, and other neurofluid drainage mechanisms rather than a single exclusively glymphatic pathway [7, 12, 13]. This multi-compartment sequence should therefore be regarded as a conceptual model for interpreting human glymphatic-related measures rather than a fully established description of all brain clearance mechanisms.

Measurable domains of glymphatic function

For human glymphatic assessment, we propose a three-domain conceptual framework for organizing human glymphatic-related measures: transport, exchange, and clearance outcome. This framework represents the authors' synthesis of physiological processes described across the glymphatic and broader neurofluid literature and is intended as an organizational tool rather than a previously validated classification. These domains are intended to organize measurable neurofluid phenomena and do not assume that each process is mediated exclusively by the glymphatic pathway. These domains describe the physiological processes of interest rather than the directness of the measurement itself. Accordingly, the physical quantity measured by each modality should be distinguished from the physiological process inferred from that measurement; the domain assignments used in this review represent proposed interpretive associations rather than experimentally validated mappings between individual biomarkers and glymphatic physiology. First, transport refers broadly to movement of fluid or solutes relevant to perivascular pathways, whereas CSF flow refers specifically to measurable CSF circulation or motion within defined CSF compartments [3, 7]. Second, exchange refers to the efficiency of CSF–ISF mixing at the perivascular and parenchymal interface, reflecting the degree to which solute transport into and out of brain tissue can occur [7, 13]. Third, clearance outcome refers to downstream retention or removal-related changes, which may be assessed through tracer kinetics or inferred indirectly from biological correlates [3]. The three domains are conceptually distinct but not mutually exclusive, and a single modality may provide information relevant to more than one domain. Accordingly, assignment of a measure to a primary domain should not be interpreted as evidence that the modality exclusively represents that domain or directly measures the corresponding glymphatic process. This domain-based framework is intended to reduce conceptual ambiguity and support more consistent interpretation across heterogeneous methods.

Key anatomical structures relevant to imaging

Several anatomical components are particularly relevant for clinical imaging and proxy interpretation. PVS are proposed to serve as important conduits for perivascular fluid movement along penetrating vessels and are visible on structural MRI when enlarged or remodeled [3, 14, 15]. Astrocytic endfeet, enriched with AQP4 water channels, form an important interface for water flux and may influence CSF–ISF exchange [2, 3]. Perivascular, venous, and meningeal lymphatic pathways may contribute to downstream fluid and solute drainage, although their relative contributions

and organization remain incompletely established [12, 13]. These structures highlight that brain clearance involves interacting neurofluid pathways rather than a single anatomically isolated process.

Main driving forces and state dependence

Perivascular fluid and solute transport is not constant and is shaped by physiological forces that vary over time. Arterial pulsatility is considered a major driver of perivascular CSF movement, and reduced vascular compliance may therefore influence transport efficiency [16, 17]. Respiration and pressure gradients contribute to CSF oscillations and can modulate fluid movement through changes in intrathoracic and intracranial pressure [16, 18]. Importantly, glymphatic-related neurofluid dynamics are brain-state dependent. Human studies demonstrate coordinated sleep-related changes in CSF and neurofluid dynamics [5, 19], whereas enhanced CSF-ISF exchange relative to wakefulness has been demonstrated most directly in preclinical studies [20]. These driving forces are central to clinical interpretation because they can act as modulators or confounders of imaging-based proxies.

Why glymphatic function is not a single measurable variable

Rather than representing a single measurable entity, glymphatic function must be inferred from complementary measures that differ in both the physiological domain they interrogate and their degree of measurement directness. Existing approaches characterize different aspects of neurofluid physiology, such as perivascular diffusion patterns, CSF flow dynamics, tracer influx and clearance kinetics, or structural remodeling of PVS [3]. Importantly, these physiological domains are not mutually exclusive, and the domain relevance of a given modality may vary according to acquisition timing, analytical approach, and the physiological variable being interpreted. These approaches range from tracer-informed measures of CSF distribution and clearance to physiology-sensitive neurofluid proxies, indirect imaging correlates, and downstream biological correlates. Moreover, these measurements may reflect overlapping glymphatic, perivascular, meningeal lymphatic, and other neurofluid clearance mechanisms rather than an exclusively glymphatic process [7, 12, 13]. As a result, measures derived from different modalities are not interchangeable and may appear inconsistent across studies even when they are each biologically valid [7]. The proposed transport–exchange–clearance framework should therefore be interpreted together with this evidence hierarchy, providing a practical basis for selecting appropriate measures, aligning methods with clinical

questions, and improving comparability across studies. The framework is intended to organize measurable physiological phenomena without assuming that their underlying mechanisms are uniquely or completely glymphatic.

Indicators and proxies for human glymphatic assessment

No single biomarker captures all aspects of glymphatic physiology. Instead, current human assessment approaches differ in both the physiological domain they interrogate and the directness of the evidence they provide. The following sections group major measures by measurement type to clarify what each metric directly measures, what glymphatic-related process may be inferred, and its clinical and translational relevance. The major glymphatic-related indicators and proxies, their proposed domain relevance, and measurement characteristics are summarized in Table 1.

Imaging-based measures

DTI-ALPS index

The diffusion tensor imaging along the perivascular spaces (DTI-ALPS) index was originally introduced by Taoka et al. as a diffusion MRI approach for evaluating directional water diffusivity along the perivascular-space direction in a specific periventricular white-matter configuration [21]. The ALPS index is derived from standard diffusion MRI by comparing directional diffusivities measured in projection- and association-fiber regions adjacent to the lateral ventricles [21]. Diffusion MRI measures probabilistic Brownian displacement of water constrained by local tissue structure rather than unidirectional bulk fluid flow [22]. Because the ALPS index is a ratio of multiple directional diffusivities, a reduced value may result from decreased numerator diffusivity, increased denominator diffusivity, or both [22]. It is currently considered an indirect diffusion-derived imaging correlate that has been investigated in relation to glymphatic-related physiology [9, 23]. As a non-invasive metric readily obtained from routine DTI datasets, ALPS has been widely applied in clinical cohorts and has demonstrated associations with aging and neurodegenerative diseases, including Alzheimer's and Parkinson's disease [23, 24]. Methodological factors also influence ALPS values and are discussed in Sect. "Diffusion MRI techniques (DTI-ALPS, IVIM)" [22, 25]. Its biological specificity remains uncertain because the ALPS index may be influenced by white-matter microstructure, fiber orientation, crossing fibers, extracellular geometry, and vascular/perivascular geometry [22, 23]. Consequently, a reduced ALPS index should not be

Table 1 Indicator–domain mapping of glymphatic assessment measures

Indicator	Physical quantity measured	Inferred physiological relevance	Evidence class	Strengths	Key limitations	Current research
DTI-ALPS index	Directional water diffusivity in defined periventricular white-matter regions derived from DTI. [21, 22]	Transport-related (indirect): localized diffusion characteristics potentially relevant to perivascular/neurofluid physiology; not a direct measure of bulk fluid transport. [22, 23]	Indirect imaging correlate	Non-invasive; routine DTI; scalable. [23]	Indirect; influenced by white-matter microstructure and sensitive to protocol, ROI definition, motion, and acquisition geometry; cross-site variability. [22, 23, 25]	Scalable diffusion-derived correlate; interpret multimodally rather than as direct transport or clearance. [22, 23]
Tracer MRI (GBCA kinetics)	Spatial and temporal GBCA distribution, parenchymal enrichment, and washout kinetics across CSF and tissue. [6, 26]	Transport / Exchange / Clearance Outcome (tracer-informed): domain relevance depends on tracer location, acquisition timing, and analytical approach; does not comprehensively measure the entire glymphatic pathway. [6, 24, 26]	Tracer-informed measure	Relatively direct tracer-informed assessment of selected neurofluid processes; approximates experimental tracer paradigms and detects delayed tracer clearance. [6, 24, 26]	Invasive/off-label for intrathecal administration; prolonged protocols; sensitive to tracer characteristics, imaging timing, individual CSF dynamics, and, for IV approaches, BBB integrity; does not comprehensively quantify glymphatic clearance. [6, 24, 26]	Tracer-informed research validation; causal and therapeutic studies. [6, 26]
BOLD–CSF coupling	Temporal coupling between global BOLD fluctuations and CSF-related signal changes on fMRI. [19, 28]	Transport / Exchange-related (physiology-sensitive proxy): coordinated neurovascular-neurofluid dynamics potentially relevant to CSF movement; not a direct measure of CSF–ISF exchange or solute clearance. [19, 28]	Physiology-sensitive neurofluid proxy	Non-invasive; uses routine fMRI datasets; state-sensitive. [19, 28]	Indirect; the magnitude, sign, and temporal lag of coupling are sensitive to motion, physiological state, preprocessing, CSF ROI definition, lag/correlation method, and analytical convention, limiting direct comparison across studies. [19, 28]	Mechanistic and state-dependent studies; combine with physiological monitoring. [19, 28]
ePVS burden	Structural burden of visible/dilated perivascular spaces assessed primarily on T2-weighted MRI, with FLAIR as a complementary sequence, quantified by regional count, volume, or rating score. [14, 15]	Transport-related (indirect structural correlate): may reflect chronic perivascular remodeling or altered fluid-handling pathways but does not directly measure fluid transport or exchange. [1, 14, 24]	Indirect imaging correlate	Highly feasible; clinically familiar; suitable for large cohorts. [14, 15, 29]	Nonspecific; influenced by vascular and other pathological processes; dependent on sequence and rating/segmentation methodology. [15, 24, 29]	Structural correlate within multimodal assessment; not a direct transport measure. [14, 15]

Table 1 (continued)

Indicator	Physical quantity measured	Inferred physiological relevance	Evidence class	Strengths	Key limitations	Current research
Sleep stage / EEG slow waves	Sleep stage and electrophysiological slow-wave activity assessed by EEG/polysomnography. [5, 19]	Exchange-related state indicator: identifies physiological states associated with CSF/neurofluid dynamics but does not directly measure CSF–ISF exchange or clearance. [5, 19, 24]	Physiological state measure	Explains state-dependent heterogeneity and enables state-controlled study designs. [5, 19, 24]	Indirect; workflow burden; medications, circadian timing, sleep disorders, and arousal may confound interpretation. [5, 24]	Control or stratify imaging studies; support state-controlled interpretation. [5, 19, 24]
Cardiorespiratory/hemodynamic measures	Respiration, CO ₂ , heart rate, blood pressure, and cardiovascular pulsatility measured using physiological monitoring. [18, 32]	Transport-related modulator/confounder: influences hemodynamics, intracranial fluid dynamics, and CSF motion but is not itself a measure of glymphatic transport, exchange, or clearance. [18, 31, 32]	Physiological modulator/confounder	Routine physiological monitoring; provides important context for interpreting neurofluid-related measurements. [18, 32]	Not a direct glymphatic measure; relationships may vary with physiological and vascular state. [18, 24, 32]	Core physiological covariates and potential confounders for imaging studies. [18, 24, 32]
CSF solute markers	Concentrations of CSF solutes or disease-related proteins, including amyloid- β , tau, and lactate. [24]	Clearance Outcome-related (downstream biological correlate): concentrations may be influenced by clearance-related processes but reflect the combined effects of production, aggregation, transport, barrier exchange, and multiple clearance pathways. [14, 24]	Downstream biological correlate	Clinically relevant endpoints; provide downstream biological context. [24, 33]	Nonspecific; cannot currently quantify glymphatic clearance in an individual; invasive sampling; sensitive to preanalytical and analytical variability. [14, 24, 33]	Integrate with imaging and physiological measures to provide downstream biological context. [14, 24]
AQP4 polarization	Perivascular localization/polarization of AQP4 at astrocytic endfeet, assessed primarily in human tissue. [36, 37]	Exchange-related (mechanistic cellular correlate): provides mechanistic information relevant to CSF–ISF water exchange but is not a direct in vivo measure of exchange or clearance. [2, 34, 35]	Mechanistic cellular correlate	Provides mechanistic context and a potential translational target. [1, 2, 34]	Mostly tissue-based in humans; limited routine in vivo assessment; alterations may reflect secondary pathology. [36, 37]	Mechanistic interpretation and future biomarker development. [34]
CSF flow MRI (4D flow / phase-contrast)	Macroscopic CSF velocity and oscillatory motion within defined CSF compartments, including velocity and waveform measures across the cardiac cycle. [63]	Transport-related (physiology-sensitive): provides information on macroscopic CSF dynamics but does not directly measure microscopic perivascular transport, CSF–ISF exchange, or solute clearance. [63]	Physiology-sensitive neurofluid measure	Non-invasive; quantitative assessment of spatially and temporally resolved CSF motion. [63]	Measures macroscopic CSF motion rather than microscopic perivascular transport; sensitive to VENC, segmentation, scanner hardware, and physiological state. [63]	Physiology-informed research assessment of CSF dynamics; integrate with complementary neurofluid measures. [63]

The “physical quantity measured” column describes the signal, structural feature, physiological variable, tracer kinetic parameter, or biological concentration directly obtained by each approach, whereas “inferred physiological relevance” indicates the glymphatic-related process to which that measurement may be interpreted as relevant. Transport, exchange, and clearance-outcome assignments represent the authors’ proposed interpretive framework and should not be considered experimentally validated mappings between individual measures and specific glymphatic processes. Domain relevance may overlap and vary according to acquisition timing, analytical approach, and physiological context

Abbreviations: *AQP4* aquaporin-4, *BOLD* blood oxygen level-dependent, *CSF* cerebrospinal fluid, *DTI-ALPS* diffusion tensor imaging along the perivascular space, *EEG* electroencephalography, *ePVS* enlarged perivascular spaces, *GBCA* gadolinium-based contrast agent, *IT* intrathecal, *MRI* magnetic resonance imaging, *ROI* region of interest, *VENC* velocity encoding

interpreted as direct evidence of reduced glymphatic transport or global glymphatic dysfunction. Recent methodological reassessment therefore suggests that DTI-ALPS is better interpreted as a structure-dependent, spatially fixed-point diffusion biomarker reflecting localized directional diffusivity rather than as a direct measure of whole-brain glymphatic transport [22]. Therefore, DTI-ALPS is best used as a scalable diffusion-derived correlate of glymphatic-related physiology and interpreted with complementary measures rather than as a direct measure of perivascular transport or global clearance [22, 23].

Contrast-enhanced tracer MRI (GBCA kinetic imaging)

Tracer-informed kinetic MRI assesses CSF distribution and clearance kinetics by tracking gadolinium-based contrast agent (GBCA) distribution over time [6]. Pioneering human work by Ringstad et al. used intrathecal gadobutrol followed by serial T1-weighted MRI to characterize brain-wide CSF tracer enrichment and clearance over time [26]. In human intrathecal tracer-MRI studies, GBCA is administered into the CSF compartment, followed by serial T1-weighted imaging over hours to days to evaluate tracer distribution, parenchymal enrichment, and washout kinetics [6, 26]. Accordingly, tracer kinetic MRI may provide information relevant to multiple domains: early tracer distribution may inform transport, parenchymal uptake may inform exchange-related processes, and delayed washout may inform clearance outcome, depending on acquisition timing and analytical approach. This approach provides relatively direct, tracer-informed information on selected components of neurofluid transport and clearance compared with diffusion or structural markers because it approximates experimental tracer paradigms and detects delayed tracer clearance in disorders involving CSF circulation dysfunction [6, 26]. However, tracer distribution and washout should not be interpreted as comprehensive measures of glymphatic clearance, because the observed kinetics depend on tracer characteristics, imaging timing, and individual CSF dynamics. Its practical and methodological limitations are discussed in Sect. " [Contrast-enhanced MRI techniques](#)" [6, 27]. Therefore, tracer kinetic MRI is best suited for tracer-informed investigation of human neurofluid pathways and mechanistic research on potential modulators or interventions rather than routine clinical screening [6].

BOLD–CSF coupling

Blood oxygen level-dependent (BOLD)–CSF coupling is a physiology-sensitive neurofluid proxy characterizing the relationship between slow fluctuations in global BOLD activity and CSF inflow/outflow signals from ventricular or

CSF regions, with potential relevance to neurofluid dynamics during low-arousal or sleep-like states [19, 28]. The seminal human study by Fultz et al. demonstrated that, during non-rapid eye movement sleep, electrophysiological slow waves are temporally coupled with hemodynamic oscillations and large CSF signal fluctuations, providing foundational evidence for coordinated neural, hemodynamic, and CSF dynamics in humans [19]. BOLD–CSF coupling can be derived from functional magnetic resonance imaging (fMRI) by examining temporal relationships between global BOLD fluctuations and CSF signal changes [19, 28]. However, the magnitude, sign, and temporal lag of the measured relationship depend on CSF ROI definition, preprocessing procedures, physiological and arousal state, and the analytical convention used to define coupling [19, 28]. Consequently, stronger inverse coupling should not be regarded as a universally interchangeable metric across studies unless these methodological factors and the direction and lag convention are comparable. Because BOLD–CSF coupling can be derived non-invasively from resting-state fMRI, it has been applied in human clinical cohorts; for example, reduced coupling has been reported in patients with depressive disorder and was associated with sleep disturbance [28]. Its interpretation remains sensitive to physiological state and methodological choices, with implementation considerations discussed in Sect. " [Functional neurofluid assessment \(BOLD–CSF coupling with fMRI ± EEG\)](#)" [28]. Therefore, BOLD–CSF coupling is most useful in mechanistic and state-dependent studies, including sleep-related and psychiatric cohorts, and is best interpreted alongside physiological monitoring or complementary imaging measures rather than as a direct measure of CSF–ISF exchange or clearance [19, 28].

Enlarged perivascular spaces (ePVS) burden

Enlarged perivascular spaces (ePVS) are structural MRI features representing dilated perivascular fluid compartments, and increased ePVS burden is a nonspecific, indirect structural imaging correlate potentially associated with altered perivascular fluid handling and clearance-pathway remodeling [1, 14]. ePVS burden can be quantified using visual or automated structural MRI approaches, with implementation considerations described in Sect. " [Structural MRI of perivascular spaces \(ePVS imaging\)](#)" [15, 29]. This clinically familiar measure supports large retrospective studies of aging and vascular brain disease [14, 15, 29]. Associations between ePVS and neurodegeneration are strengthened by reports linking ePVS burden to CSF measures related to AQP4 and tau, suggesting relevance to clearance-related pathology [1, 14]. However, ePVS is not specific to glymphatic dysfunction and may also reflect small vessel

disease, amyloid-related vasculopathy, inflammation, or generalized tissue remodeling, while subjective rating variability remains a limitation [15, 24, 29]. Therefore, ePVS is best used as a structural imaging correlate within multimodal assessment frameworks rather than a direct measure of transport or exchange dynamics [14].

Physiological state measures

Sleep stage and electroencephalogram (EEG) slow-wave activity

Sleep stage, particularly slow-wave sleep (N3), is a key physiological state associated with glymphatic-related neurofluid dynamics, while EEG slow-wave activity identifies states associated with CSF signal fluctuations rather than directly measuring CSF–ISF exchange or fluid flow [5, 19, 20, 24]. Sleep stage is assessed using polysomnography or EEG staging, and in some studies EEG–fMRI directly links slow-wave events with CSF-related signal changes [19, 24]. This physiological state measure is important because state dependence strongly influences imaging proxies and partly explains variability across human studies, while evidence suggests sleep disruption impairs clearance-related physiology and contributes to unfavorable biomarker profiles [24]. Sleep metrics can also be used to stratify participants and support state-controlled interpretation of glymphatic-related imaging measures [24, 30]. However, limitations include indirectness, limited feasibility in routine clinical practice, and confounding by medications, stress, circadian timing, and sleep disorders such as apnea affecting both sleep architecture and CSF dynamics [24]. Overall, sleep-stage measures are best used to improve interpretation of imaging biomarkers and support state-controlled clinical protocols rather than as direct measures of glymphatic function [24].

Cardiorespiratory and hemodynamic surrogates

Physiological measures including respiration, CO₂ dynamics, heart rate, blood pressure, and pulsatility provide mechanistic context as biological modulators of glymphatic-related transport because they influence intracranial pressure gradients and CSF motion; they may also act as confounders when they vary systematically across participants, study groups, or acquisition conditions [18, 31, 32]. These variables are obtained through routine monitoring, including ECG or pulse oximetry, respiratory belts, and capnography, and should ideally be recorded alongside imaging biomarkers to characterize physiological state and reduce confounding [18, 32]. Their main strength is explaining within- and between-subject variability and improving reproducibility by distinguishing disease-related changes

from arousal- or breathing-related effects during imaging [18, 32]. However, these measures are not direct indicators of glymphatic exchange or clearance, and their relationship with transport may be altered by chronic vascular injury, arterial stiffness, or hypertension that remodel perivascular pathways [24]. Therefore, cardiorespiratory metrics are best treated as important physiological covariates and potential confounder controls that strengthen interpretation of imaging biomarkers [18, 24, 32].

Downstream biological correlates

CSF solute biomarkers

CSF solute markers represent downstream biological correlates that may be influenced by clearance processes, including measures of metabolic byproducts and disease-related proteins such as amyloid- β , tau, and lactate [24]. These biomarkers should not be interpreted as direct measures of glymphatic clearance because their concentrations reflect the combined effects of multiple biological processes and cannot currently quantify glymphatic clearance in an individual patient. These biomarkers are obtained through lumbar puncture with standardized immunoassays and can be complemented by amyloid or tau positron emission tomography (PET) to assess long-term accumulation patterns [24]. Their major strength is direct clinical relevance, particularly in neurodegenerative diseases where CSF biomarkers support diagnosis and prognosis, providing clinically relevant downstream information that may be interpreted alongside neurofluid measurements [24]. Standardized preanalytical and analytical procedures are also essential for improving the reliability and interpretation of CSF biomarker measurements, particularly for amyloid- β analysis [33]. However, interpretation is limited by low specificity because biomarker concentrations also depend on production, aggregation, barrier transport, and alternative clearance pathways, including meningeal lymphatic drainage [24]. In addition, CSF sampling is invasive. Therefore, CSF solute markers are best integrated with imaging and physiological measures to provide downstream biological context rather than being interpreted as specific indicators of glymphatic clearance [14].

Mechanistic cellular indicator

AQP4 polarization

AQP4 polarization refers to the preferential localization of AQP4 water channels at astrocytic endfeet surrounding cerebral vessels, forming a key molecular interface that supports CSF–ISF water exchange and influences glymphatic-related

exchange efficiency [2, 34, 35]. In humans, direct assessment of AQP4 expression and perivascular localization has primarily relied on histopathological analysis of surgical or postmortem brain tissue, while routine *in vivo* quantification remains limited, making this indicator primarily translational [36, 37]. Its major strength is mechanistic value, as experimental evidence links loss or altered polarization of perivascular AQP4 with impaired glymphatic-related fluid transport and clearance, while AQP4 has also been investigated as a potential therapeutic target [2, 34, 35]. However, clinical implementation remains limited by feasibility, and AQP4 expression or polarization may be altered across neurological and neurodegenerative disease states, complicating causal interpretation across heterogeneous patient populations [2, 35]. Therefore, AQP4 polarization is best regarded as a mechanistic and translational indicator rather than a routine clinical biomarker [35–37].

Modulating factors and confounders in human glymphatic assessment

Interpretation of glymphatic-related measures requires consideration of physiological state and systemic conditions, as the same measure may vary even without a corresponding change in the underlying neurofluid process of interest. In this review, modulators are defined as physiological, behavioral, or systemic factors that may biologically influence glymphatic-related processes or their measured proxies, whereas confounders are factors that may bias comparisons or interpretation when they differ systematically across participants, groups, or acquisition conditions. Importantly, the same variable may serve both roles depending on the study context. The evidence supporting individual modulators varies across direct experimental animal studies, human tracer studies, human physiological or imaging-proxy studies, and observational associations; these evidence sources are distinguished throughout this section and summarized in Table 2. Mechanistic effects demonstrated in animal models are therefore not assumed to represent established human glymphatic physiology. Below, key modulators are grouped by category to highlight factors that commonly contribute to variability across diffusion-, functional-, and tracer-based assessments.

Behavioral and state-dependent modulators

Sleep–wakefulness / arousal

This factor modulates diffusion-, functional-, and tracer-based glymphatic indicators by altering physiological state [1, 5, 31, 38]. Evidence is derived from both preclinical

mechanistic studies and human neuroimaging studies. Foundational preclinical work by Xie et al. demonstrated in mice that natural sleep or anesthesia was associated with expansion of the interstitial space, increased CSF–ISF exchange, and enhanced β -amyloid clearance compared with wakefulness [20]. Within the proposed framework, sleep–wake state is assigned primarily to exchange-related physiology, with secondary relevance to transport [5, 19, 20]. Consistent with these preclinical findings, human studies have demonstrated sleep-related changes in CSF and glymphatic-relevant neurofluid signals rather than direct measurement of glymphatic clearance. Simultaneous EEG–fMRI studies show NREM slow waves are time-locked to coordinated CSF signal fluctuations [19, 39], with more recent work also identifying relationships with spindle activity [39]. A recent systematic review summarized evidence linking sleep disruption with altered CSF-related and glymphatic-relevant measures, reinforcing sleep state as a major source of heterogeneity in human glymphatic proxies [5].

Circadian timing

Circadian timing may influence glymphatic-related measures, although direct mechanistic evidence for circadian regulation of glymphatic transport and clearance is derived primarily from preclinical studies [31, 34]. Animal studies have demonstrated daily oscillations in glymphatic influx and clearance and circadian variation in AQP4 localization [34]. These findings suggest that transport and exchange may vary with circadian phase, but their direct translation to human glymphatic function remains incompletely established. In humans, however, one diffusion-MRI study found no significant time-of-day dependence in glymphatic-related diffusion measures during the awake state under the sensitivity of the applied methods [9]. Nevertheless, recording acquisition time remains useful for reproducibility and comparison across studies [9, 34].

Physiological driving-force modulators

Respiration and CO₂ dynamics

Respiration and CO₂ dynamics modulate functional MRI coupling metrics and CSF flow-related measures by altering cerebral hemodynamics and intracranial pressure oscillations [10, 32]. Human physiological and imaging studies support respiratory and CO₂-related modulation of cerebral hemodynamics and CSF movement [10, 32]. Mechanistically, respiratory patterns and arterial CO₂ can alter cerebral hemodynamics and CSF motion; hypercapnia induces cerebral vasodilation and increased cerebral blood flow, while CO₂-driven changes in cerebral blood volume have been

Table 2 Factors relevant to human glymphatic assessment: proposed physiological relevance, evidence source, and recommended controls

Factor	Proposed / inferred physiological relevance	Level/source of evidence	Reported association	Measures to record	Recommended control	Most affected modalities
Sleep–wake / arousal [5, 19, 20, 38, 39]	Primarily exchange-related; secondarily transport-related	Animal experimental + human physiological/imaging proxy + human observational imaging association	Sleep is associated with state-dependent changes in CSF-related and glymphatic-relevant measures; direct enhancement of CSF–ISF exchange and clearance is supported mainly by preclinical experiments.	Sleep duration/quality; vigilance/drowsiness during scan	Standardize pre-scan sleep; record drowsiness; avoid unplanned sleep deprivation	BOLD–CSF, ALPS, Flow, Tracer
Circadian timing [9, 34]	Transport- and exchange-related	Animal experimental + human imaging-proxy association	Circadian variation in glymphatic influx and clearance has been demonstrated preclinically, while human diffusion-MRI studies have investigated age- and time-of-day-related variation in glymphatic-related proxies.	Time of day; chronotype (optional)	Fix scan window or adjust statistically	ALPS, BOLD–CSF, Flow, Tracer
Respiration / CO ₂ [31, 32]	Primarily transport-related; secondarily exchange-related	Human physiological/imaging experimental evidence	Respiration and CO ₂ manipulation are associated with changes in cerebral hemodynamics and CSF-related flow signals; these changes do not establish altered glymphatic clearance.	Respiratory rate/pattern; end-tidal CO ₂ (if feasible)	Monitor respiration; use consistent instructions; avoid uncontrolled hyperventilation	BOLD–CSF, Flow, Tracer
Arterial/cardiac pulsatility [17, 31, 40, 41]	Primarily transport-related; secondarily exchange-related	Animal experimental + human physiological/imaging evidence	Arterial pulsatility is associated with perivascular CSF-related motion, while direct mechanistic evidence linking pulsatility to glymphatic clearance is primarily preclinical.	Heart rate/rhythm; blood pressure; stiffness proxies (if available)	Record blood pressure/heart rate; note arrhythmia; maintain stable resting conditions	Flow, Tracer, ALPS
Cerebral autoregulation [42–44]	Transport-related physiology	Human physiological evidence + observational clinical association	Cerebral autoregulation influences cerebral perfusion and vascular dynamics, but a direct effect on human glymphatic transport has not been established.	Blood pressure variability; vascular disease markers	Record vascular comorbidity; avoid extreme blood pressure states	Flow, BOLD–CSF, Tracer
Vascular stiffness [16, 30, 41]	Primarily transport-related; secondarily structure-related	Primarily mechanistic/preclinical + indirect human vascular association	Reduced vascular compliance may alter pulsatility relevant to perivascular fluid dynamics, but direct effects of vascular stiffness on human glymphatic transport remain unestablished.	Age; blood pressure; vascular risk profile	Stratify by age/risk; report antihypertensive use	ALPS, Tracer, ePVS
Hypertension [30, 41]	Primarily structure-related; secondarily transport- and downstream-outcome-related	Animal experimental + indirect human observational/imaging evidence	Hypertension is associated with vascular and perivascular abnormalities in humans, while impaired glymphatic transport has been demonstrated directly in a hypertensive rat model.	Hypertension diagnosis; blood pressure control; medications	Standardize blood pressure measurement; adjust for hypertension	ePVS, ALPS, Tracer, CSF markers
Body posture [1, 45–47]	Transport-related	Human physiological/imaging evidence + preclinical evidence	Body position is associated with changes in human CSF and neurofluid dynamics; evidence for posture-dependent enhancement of glymphatic transport or clearance is primarily preclinical.	Pre-scan posture; time supine	Standardize/document posture before imaging; maintain consistent positioning during acquisition	Flow, Tracer
Microgravity / spaceflight [3, 48]	Transport- and structure-related	Human observational imaging evidence	Long-duration spaceflight is associated with changes in CSF spaces and PVS-related imaging measures, but these findings do not directly demonstrate altered glymphatic clearance.	Exposure duration	Report mission/exposure history	ePVS, Flow
Physical exercise [49, 50]	Transport- and exchange-related	Human interventional tracer-informed imaging + animal experimental evidence	Long-term exercise has been associated with increased putative glymphatic-related tracer influx in humans, while enhanced glymphatic clearance has been reported in preclinical mouse studies.	Same-day exercise; habitual activity	Record same-day exercise and habitual activity; prespecify exercise restrictions when relevant to the study design.	ALPS, Tracer, BOLD–CSF

Table 2 (continued)

Factor	Proposed / inferred physiological relevance	Level/source of evidence	Reported association	Measures to record	Recommended control	Most affected modalities
Stress [51, 52]	Exchange- and transport-related	Animal experimental evidence; limited direct human evidence	Chronic stress has been associated with reduced glymphatic transport and altered AQP4-related processes in animal models; comparable direct human evidence remains limited.	Stress scales (optional); cortisol proxy (if available)	Record major stressors; avoid acute stress manipulation unless planned	BOLD–CSF, CSF markers
Alcohol [1, 3, 53, 54]	Transport- and exchange-related	Primarily animal experimental evidence + indirect human observational/imaging evidence	Dose-dependent effects of alcohol on tracer-defined glymphatic-related transport have been reported primarily in pre-clinical studies, whereas human evidence remains indirect.	Recent intake; alcohol use history	Standardize abstinence window; record dose/time	ALPS, BOLD–CSF, ePVS, CSF markers
Pharmacological agents [8, 35]	Transport- and/or exchange-related (agent-specific)	Heterogeneous: mainly preclinical mechanistic + limited human physiological evidence	Pharmacological effects on glymphatic-related and neurofluid measurements appear agent-, dose-, and state-dependent, and no consistent directional effect has been established in humans.	Caffeine; sedatives; stimulants; neuroactive medications	Use pre-scan restriction windows; document dose/time	ALPS, BOLD–CSF, Flow, Tracer
Anesthetics [1, 55]	Transport-related; potential secondary exchange relevance	Primarily animal experimental evidence	Different anesthetic conditions are associated with different tracer-defined glymphatic-related transport patterns in animal studies; equivalent effects on human glymphatic clearance remain unestablished.	Agent; dose; timing; vitals	Harmonize protocols; avoid mixing agents across groups	Tracer, Flow, ALPS
Aging [1, 9, 30, 56, 58]	Transport-, exchange-, and downstream-outcome-related	Human imaging associations + preclinical mechanistic evidence	Aging is associated with changes across several human glymphatic-related imaging measures, although these measures do not establish a uniform age-related reduction in glymphatic function.	Age; vascular risk	Age-match; stratify; report comorbidities	All modalities
Sex and hormonal status [8, 59]	Potential association with neurofluid/glymphatic-related physiology	Limited preclinical + emerging human observational/imaging evidence	Sex and hormonal status have been proposed as potential contributors to glymphatic-related physiology, but the direction and magnitude of their effects on human glymphatic function remain uncertain.	Sex; menopausal/hormonal status (if relevant)	Stratify; report hormonal status/HRT if applicable	BOLD–CSF, ALPS, Tracer

Abbreviations: *AQP4* aquaporin-4, *ALPS* diffusion tensor imaging along the perivascular space, *BOLD* blood oxygen level-dependent, *CO₂* carbon dioxide, *CSF* cerebrospinal fluid, *ePVS* enlarged perivascular spaces, *HRT* hormone replacement therapy

Factors are listed according to their potential biological role as modulators of glymphatic-related physiology or measured proxies. The same factor may also act as a confounder depending on study design. Evidence sources are classified as animal experimental, human tracer-informed, human physiological/imaging proxy, or human observational association, with combinations indicated where applicable. Reported associations or effects refer to the specific quantity measured in the cited studies and should not be interpreted as established changes in human glymphatic transport, exchange, or clearance when evidence is indirect, observational, or derived primarily from animal experiments

associated with bidirectional macroscopic CSF movement [10, 32]. Within the proposed framework, respiration and CO₂ dynamics are assigned primarily to transport-related physiology, with secondary relevance to exchange [32]. Respiratory and CO₂ manipulations can alter CSF-related signals, although these changes should not be interpreted as direct evidence of altered glymphatic clearance [31, 32]. Therefore, breathing parameters and CO₂ status should be monitored when interpreting state-sensitive metrics [32].

Arterial and cardiovascular pulsatility

Cardiovascular pulsatility is considered an important driver of perivascular fluid movement, based largely on mechanistic and preclinical evidence [17, 31, 40, 41]. Mechanistically, arterial pressure waves induce perivascular wall motion that may facilitate CSF movement along periarterial spaces, whereas reduced pulsatile energy associated with arterial stiffness or altered cardiovascular function may

weaken this driving force [16, 17, 31]. Within the proposed framework, arterial and cardiovascular pulsatility are assigned primarily to transport-related physiology, with potential secondary relevance to exchange because altered flow may influence solute mixing. Preclinical studies support an association between altered vascular dynamics and impaired perivascular fluid or solute transport [40, 41]. Emerging human physiological MRI evidence indicates cardiac-cycle dependence of perivascular CSF-related diffusion signals, although this does not directly quantify CSF-ISF exchange or whole-brain clearance [17].

Cerebral autoregulation

Autoregulatory control of cerebral blood flow may influence glymphatic-related proxies indirectly through its effects on cerebral perfusion and intracranial fluid dynamics [42, 43]. Human physiological and clinical studies demonstrate alterations in cerebral autoregulation in relation to vascular conditions, including small vessel disease, although direct effects on glymphatic transport have not been established [42–44]. Mechanistically, intact autoregulation buffers systemic blood pressure fluctuations and maintains stable cerebral perfusion, whereas impaired autoregulation may increase variability in hemodynamic signals; whether this translates into measurable changes in CSF-related dynamics remains uncertain [42–44]. Accordingly, autoregulatory status should be considered a physiological contributor or confounder when interpreting neurofluid-related measurements rather than a direct measure of glymphatic function.

Vascular and hemodynamic health modulators

Vascular stiffness

Arterial stiffness, often age-related, may influence glymphatic-related diffusion and tracer-kinetic measures through altered vascular pulsatility [16, 30]. Mechanistically, reduced vascular compliance may dampen arterial pulsations that contribute to perivascular CSF movement [16]. Transport is therefore a potentially affected domain, with secondary structural effects because vascular stiffness and reduced pulsatility have been associated with perivascular alterations [16, 30]. Evidence linking vascular stiffness specifically to impaired glymphatic transport is derived from preclinical mechanistic findings and indirect human associations; therefore, vascular stiffness should be considered an important vascular covariate rather than a direct indicator of glymphatic dysfunction [16, 30, 41].

Hypertension

Chronic hypertension is associated with alterations in structural and other glymphatic-related imaging measures and

may influence perivascular pathways through small-vessel injury [30, 41]. Mechanistically, hypertensive remodeling, including arterial wall thickening and blood–brain barrier disruption, has been proposed to impair perivascular CSF movement and waste transport, with direct mechanistic support derived largely from preclinical studies [30, 41]. In humans, hypertension is associated with vascular and perivascular imaging abnormalities, but these findings do not independently establish impaired glymphatic clearance. Accordingly, hypertension should be recorded and controlled as an important vascular comorbidity when interpreting ePVS, diffusion-derived, tracer-informed, and downstream biological measures [30].

Body mechanics and gravity modulators

Body posture

Body and head position modulate transport-related glymphatic measures by altering hydrostatic gradients and drainage pathways [1, 45]. Human physiological and imaging studies demonstrate posture-dependent changes in CSF dynamics, with recumbency, including supine positions, supporting different venous and CSF flow patterns compared with upright posture [45]. Within the proposed framework, body posture is considered primarily relevant to transport-related physiology, with potential secondary relevance to exchange because CSF flow between the brain and spinal canal differs between supine and upright positions [45]. Preclinical studies further suggest that body posture may influence glymphatic transport and clearance, with rodent experiments demonstrating more efficient glymphatic transport in the lateral position than in the prone position [46]. However, direct evidence that lateral positioning enhances glymphatic clearance in humans remains limited. Posture should therefore be standardized or documented because it represents a potentially important source of variability in neurofluid measurements [45, 47].

Microgravity and spaceflight

Microgravity during spaceflight is associated with alterations in structural and transport-related neurofluid indicators, including PVS changes after long-duration missions, based on human astronaut and cosmonaut imaging studies [3, 48]. The absence of gravity produces cephalad fluid shifts and alters venous and CSF pressure relationships, which may influence intracranial CSF dynamics and perivascular fluid physiology [48]. Structure and transport are therefore potentially affected, as reflected by observed changes in PVS and CSF-related imaging measures [48]. However, these human imaging findings do not directly demonstrate reduced

glymphatic clearance capacity. Although uncommon clinically, these findings highlight the potential importance of gravitational and postural forces in human neurofluid physiology [48].

Lifestyle and systemic condition modulators

Physical exercise

Physical activity has been associated with changes in contrast-enhanced MRI measures of putative glymphatic influx and meningeal lymphatic flow in humans [49]. Evidence includes both human imaging studies and supporting preclinical experiments [49, 50]. The mechanisms underlying these associations remain incompletely established. In the human intervention study, long-term exercise was accompanied by changes in inflammation- and immune-related molecular profiles associated with putative glymphatic or meningeal lymphatic measures [49]. Within the proposed framework, physical exercise is considered relevant to both transport- and exchange-related physiology, with long-term aerobic training associated with increased putative glymphatic tracer influx in humans [49], while voluntary exercise has been shown to enhance glymphatic clearance in preclinical mouse studies [50]. Long-term regular exercise has been associated with favorable changes in putative glymphatic-related imaging measures in humans [49]. Physical activity should therefore be documented and considered during interpretation, particularly in interventional studies.

Stress

Stress, particularly chronic stress, has been implicated as a potential modulator of glymphatic-related physiology, although direct mechanistic evidence is derived primarily from preclinical studies [51, 52]. In animal models, glucocorticoid signaling has been associated with altered astroglial AQP4 polarization and neurovascular function, and experimental chronic stress or corticosterone exposure has been reported to reduce glymphatic transport and alter perivascular AQP4 expression [52]. These findings suggest potential effects on exchange and transport, with possible downstream consequences for waste accumulation and neuroinflammation [52]. However, comparable direct evidence of stress-induced glymphatic impairment in humans remains limited; therefore, these preclinical findings should not be directly extrapolated to human glymphatic function.

Alcohol

Alcohol intake may influence glymphatic-related physiology in a dose-dependent manner, although the strongest

mechanistic evidence for a biphasic effect is derived from preclinical studies [3, 53, 54]. In animal models, low-dose acute alcohol exposure has been associated with increased CSF tracer influx along paravascular pathways, whereas intermediate or high acute doses and prolonged intermediate-dose exposure reduce tracer transport [54]. Mechanistically, these effects may involve changes in vascular physiology, sleep, inflammation, and AQP4-related processes [3, 53]. Human studies provide more indirect evidence through structural and diffusion-related findings associated with chronic alcohol exposure, rather than direct demonstration of altered glymphatic clearance [1, 53]. Alcohol intake should therefore be documented, including dose and timing, when interpreting glymphatic-related imaging measures.

Pharmacological and clinical-state modulators

Pharmacological modulators

Medications and drugs may influence glymphatic-related physiology through effects on sleep and arousal, vascular physiology, and AQP4-related water transport [35]. Evidence is heterogeneous and includes preclinical mechanistic studies, human physiological studies, and indirect observations rather than consistent direct measurements of human glymphatic function. Stimulants such as caffeine and sedative or sleep-promoting agents may alter neurofluid-related measurements through their effects on cerebral vascular tone, arousal, and sleep physiology [8, 35]. Mechanistically, agents affecting arousal, vascular tone or pulsatility, noradrenergic signaling, or astrocytic AQP4 function may influence processes relevant to CSF movement and CSF–ISF exchange [8, 35]. However, the direction and magnitude of these effects are agent-, dose-, and state-dependent and should not be generalized across pharmacological classes. Pre-scan exposure, including over-the-counter medications, should therefore be documented because it may influence measured metrics.

Anesthetics

Anesthetic agents substantially influence glymphatic-related measurements in preclinical studies and represent an important confounder in translational research [1, 55]. Most direct evidence concerning anesthesia-induced changes in glymphatic transport and clearance derives from animal tracer experiments. Anesthesia alters neural activity, cardiovascular physiology, and arousal state, all of which can influence CSF and perivascular fluid dynamics [1, 55]. In rodent studies, anesthetic regimens produce different effects on tracer influx and glymphatic-related transport; ketamine/xylazine and other agents show distinct patterns

depending on anesthetic type and physiological conditions [1, 55]. These findings indicate that transport and potentially exchange-related processes are sensitive to anesthetic state [1, 30]. However, these preclinical findings should not be assumed to represent equivalent changes in human glymphatic clearance. Anesthetic agent, dose, timing, and physiological conditions should therefore be standardized and reported in translational studies.

Biological and demographic modulators

Aging

Aging is a major biological factor associated with changes in glymphatic-related measures across multiple modalities [1, 30, 56]. Evidence includes human imaging studies together with preclinical mechanistic studies [1, 30, 56]. Mechanistically, aging is associated with arterial stiffening, altered cardiovascular pulsatility, and astroglial changes that may influence CSF circulation, perivascular transport, and waste-clearance pathways [31, 56, 57]. Transport, exchange, and downstream outcomes may therefore be affected, while human studies have reported age-related changes in tracer-informed, diffusion-derived, and other neurofluid-related measures [56, 58]. Human diffusion MRI studies have reported age-related changes in glymphatic-related measures, although the direction of these changes varies according to the metric examined [9]. Although lifestyle and vascular comorbidities contribute to inter-individual variability, age remains a major source of systematic variation in glymphatic-related measurements [56, 57].

Sex and hormonal status

Sex and hormonal status may influence glymphatic-related physiology, but current evidence remains limited and heterogeneous [8, 59]. Available evidence includes preclinical mechanistic findings and emerging human observational or imaging associations rather than definitive evidence of sex-specific differences in human glymphatic clearance [8]. Hormonal influences on vascular physiology and astroglial AQP4-related processes have been proposed as potential mechanisms linking sex and hormonal status to neurofluid dynamics [8, 59]. However, the magnitude, direction, and clinical significance of sex- or menopause-related differences in human glymphatic function have not been established [8]. Therefore, sex and, when relevant, menopausal or hormonal status should be recorded and considered in study design and analysis, without assuming a predefined direction of effect [8].

Measurement strategies for glymphatic assessment

Human approaches to glymphatic assessment differ substantially in practicality, invasiveness, acquisition burden, accessibility, and reproducibility across scanners and sites. Because the physiological interpretation and evidence class of individual measures are discussed in Sect. "Indicators and Proxies for Human Glymphatic Assessment", this section focuses primarily on their acquisition requirements, implementation constraints, and reproducibility. The major practical characteristics of these strategies are summarized in Table 3.

Non-invasive MRI-based techniques

Non-invasive MRI methods can be incorporated into human research without exogenous intrathecal tracers, but their implementation varies considerably in acquisition time, technical complexity, and sensitivity to protocol and physiological state.

Structural MRI of perivascular spaces (ePVS imaging)

Structural MRI primarily visualizes ePVS using T2-weighted imaging, whereas FLAIR serves as a complementary sequence for distinguishing PVS from surrounding white matter lesions and other differential diagnoses [15]. Visual rating scales or automated segmentation methods can be used to quantify regional PVS burden. Automated approaches have shown good cross-scanner reproducibility, whereas manual ratings remain susceptible to inter-rater variability [29]. Practical implementation therefore requires standardized sequence selection, regional definitions, rating or segmentation procedures, and predefined thresholds. Because structural MRI is widely available and imposes minimal additional acquisition burden, ePVS imaging is readily scalable for research cohorts [15, 29].

Diffusion MRI techniques (DTI-ALPS, IVIM)

Diffusion MRI can be incorporated into standard research MRI protocols and shows good repeatability under standardized acquisition conditions, with reported intraclass correlation values around 0.83 for ALPS in some studies [25]. Reproducibility depends strongly on acquisition and analysis conditions, including head position, imaging-plane orientation, diffusion-encoding directions, echo time, and ROI definition [25]. Standardized imaging planes, predefined ROI placement, motion-quality criteria, and harmonized reconstruction procedures are therefore important for

cross-site comparison. IVIM requires multiple low b -values and longer acquisitions than conventional DTI, together with more complex model fitting [60]. Overall, diffusion MRI is scalable and non-invasive, but reliable multicenter implementation requires protocol harmonization, standardized acquisition, and quality control [25, 60].

Perfusion and water-exchange MRI (arterial spin labeling, ASL)

Arterial spin labeling (ASL) is a non-invasive MRI technique that magnetically labels arterial blood water as an endogenous tracer. Conventional ASL primarily measures cerebral blood flow and should not be interpreted as a direct measure of glymphatic transport or water exchange [6]. Specialized ASL approaches can extend this principle to investigate water exchange across the blood-CSF barrier using modified acquisition and modeling strategies [61]. For example, Peticlerc et al. used multiple pseudo-continuous ASL acquisitions with varying labeling durations and post-labeling delays, combined with a multi-echo 3D-GRASE readout and long echo times to detect labeled water transported from blood into CSF [61]. A compartmental dynamic model was then used to estimate blood-to-CSF water exchange [61]. These exchange-sensitive approaches require specialized acquisition and analysis and currently remain research-oriented. Their interpretation should therefore distinguish conventional perfusion-related ASL signals from specifically modeled water-exchange measurements [6, 61].

Metabolic/exchange MRI (CEST-based, e.g., DGE)

Chemical exchange saturation transfer (CEST)-based approaches, including dynamic glucose-enhanced (DGE) MRI, use IV D-glucose and repeated imaging to characterize glucose-related signal changes in tissue and CSF [62]. Acquisition commonly exceeds 10 min and requires detection of relatively small signal changes, increasing susceptibility to motion and physiological variability [62]. Careful acquisition and motion control are therefore important because motion-related signal changes may be comparable in magnitude to the true CEST effect [62]. Consequently, DGE-CEST currently remains primarily research-oriented and requires further methodological validation for broader human application [62].

CSF flow dynamics MRI (4D flow MRI)

4D flow MRI extends phase-contrast imaging to quantify three-dimensional CSF velocity and oscillatory motion

throughout the cardiac cycle within macroscopic CSF compartments, such as the ventricles, cerebral aqueduct, and subarachnoid spaces [63]. These measurements characterize bulk or macroscopic CSF dynamics and should not be interpreted as direct visualization of microscopic perivascular or interstitial fluid transport within the glymphatic pathway [63]. Typical acquisitions require approximately 10–20 min or longer and substantial post-processing. Measurement of very low CSF velocities requires optimized velocity encoding, including low- or dual-VENC approaches, while quantitative outputs are sensitive to scanner hardware, VENC selection, segmentation, and physiological state [63]. Standardized acquisition and post-processing procedures are therefore necessary for reproducible cross-site implementation, and the relatively high acquisition and analysis burden currently limits widespread routine use. Accordingly, 4D flow MRI is best interpreted as a physiology-sensitive measure of macroscopic CSF motion that may provide complementary information relevant to neurofluid transport, rather than as a direct measure of microscopic glymphatic transport or CSF–ISF exchange [63].

Functional neurofluid assessment (BOLD–CSF coupling with fMRI $\pm$ EEG)

Simultaneous fMRI and EEG/polysomnographic acquisition enables state-resolved assessment of neurofluid-related signals such as BOLD–CSF coupling while characterizing vigilance and sleep state [19, 39]. BOLD–CSF coupling can also be derived from resting-state fMRI without simultaneous EEG [28]. Simultaneous fMRI–EEG acquisition requires additional EEG setup, MRI-compatible equipment, and specialized expertise. Measured BOLD–CSF coupling estimates are sensitive to participant arousal, physiological fluctuations, motion, CSF ROI definition, lag/correlation methods, and preprocessing choices [19, 28, 39]. These factors can affect the magnitude, sign, and estimated temporal lag of the BOLD–CSF relationship, limiting direct comparison of coupling values across studies using different analytical pipelines. Standardized instructions, physiological monitoring, motion-control procedures, and harmonized analytical methods are therefore essential for between-study comparison.

Contrast-enhanced MRI techniques

Contrast-enhanced MRI requires additional safety, timing, and operational considerations compared with non-invasive MRI. Practical implementation depends on the administration route, serial imaging schedule, tracer penetration, physiological state, and standardized timing.

Table 3 Comparison of measurement strategies by modality for glymphatic assessment

Modality	Invasiveness	Scan burden	Output	Reproducibility / major sources of variability	Current research / translational applicability	Key confounders
Structural MRI (ePVS imaging)	None	Short single-visit structural MRI acquisition	PVS score/count/volume (regional) [15, 29]	Influenced by sequence choice, segmentation/rater method, and scoring thresholds [15, 29]	Scalable structural correlate; routine MRI acquisition, but not validated as an individual-level glymphatic screening measure [15]	Vascular disease/age; scan protocol; rater/thresholds [14, 15]
Diffusion MRI (DTI-ALPS, IVIM)	None	Single-visit acquisition; IVIM generally requires longer acquisition than conventional DTI [60]	ALPS index; IVIM parameters [6, 25, 60]	Sensitive to acquisition geometry, ROI definition, b-values, motion, and TE; cross-protocol harmonization remains important [25, 60]	Scalable research correlate (ALPS); specialized research (IVIM); not validated for individual-level glymphatic screening [25, 60]	Motion; head position; time-of-day/state; protocol harmonization [9, 25]
Perfusion / water-exchange MRI (ASL)	None	Conventional ASL: short single-visit acquisition; exchange-sensitive ASL requires specialized multi-delay/multi-echo acquisitions [61]	Conventional ASL: cerebral blood flow; specialized exchange-sensitive ASL: modeled blood-to-CSF water exchange [6, 61]	Dependent on labeling duration, post-labeling delay, echo time, SNR, physiological state, acquisition parameters, and modeling approach [6, 61]	Conventional ASL is established for perfusion assessment; exchange-sensitive ASL remains a specialized research approach for investigating blood-to-CSF water exchange [6, 61]	CO ₂ /respiration; blood pressure; arousal; motion [6, 31, 34]
Metabolic/exchange MRI (CEST/DGE)	IV glucose	Often > 10 min; single visit	DGE/CEST exchange-sensitive signal metrics [62]	Limited by small signal changes, motion sensitivity, physiological variability, and nonstandardized acquisition/analysis [62]	Specialized research assessment; insufficiently standardized for routine application [62]	Motion; timing; glucose dosing/physiology; protocol variability [62]
CSF flow MRI (4D flow / phase-contrast)	None	~10–20+ min; single visit	Macroscopic CSF velocity, stroke volume, and waveform metrics within defined ventricular/subarachnoid CSF compartments; VENC-dependent [63]	Sensitive to VENC selection, segmentation, hardware, and measurement of very low velocities [63]	Physiology-informed assessment of macroscopic CSF dynamics; does not directly measure microscopic perivascular or interstitial glymphatic transport [63]	Respiration/CO ₂ ; cardiac rhythm; blood pressure/pulsatility; motion [18, 63]
Multimodal functional assessment (fMRI + EEG)	None	Single-visit fMRI acquisition (+ EEG setup when used)	BOLD–CSF coupling magnitude/sign and temporal lag, defined according to the specified ROI and analytical convention; state metrics [28]	Coupling magnitude, sign, and estimated lag are influenced by arousal/physiological state, motion, physiological-noise correction, CSF ROI definition, preprocessing, lag/correlation method, and analytical convention [19, 28]	Mechanistic/state-dependent research assessment; individual-level clinical utility not established [28]	Arousal/sleep; respiration/CO ₂ ; motion; time-of-day; preprocessing choices [5, 28, 34]

Table 3 (continued)

Modality	Invasiveness	Scan burden	Output	Reproducibility / major sources of variability	Current research / translational applicability	Key confounders
IV GBCA-enhanced MRI (delayed T1/T1-mapping)	IV	Multi-timepoint imaging over hours; number and timing of acquisitions are protocol-dependent [6]	Subtle enhancement; delayed T1/T1-map kinetics [6]	Sensitive to imaging timing, BBB permeability, and protocol differences [6]	Tracer-informed research assessment with translational potential; limited by low tracer penetration and prolonged protocols [6]	Timing; BBB status; blood pressure; sleep–wake state; motion [6, 34]
IT GBCA-enhanced MRI (intrathecal tracer kinetics)	Intrathecal (LP; off-label)	Serial multi-timepoint imaging over 24–48 h; timing is protocol-dependent [6, 26]	Uptake/influx + washout kinetics (time-to-peak, half-life/AUC), regional kinetics [6, 26]	Influenced by tracer dose, imaging timing, CSF dynamics, and underlying disease state [6, 27]	Tracer-informed validation/research assessment; specialized and not suitable for routine screening [6, 27]	Tracer dose; imaging timing; CSF dynamics; underlying disease state; safety considerations [6, 27]
Radionuclide cisternography	Intrathecal + radiation	Serial imaging over several hours; timing depends on tracer and protocol [64]	CSF distribution/circulation patterns; ventricular tracer distribution [64]	Dependent on tracer administration and serial imaging timing [64]	Specialized assessment of CSF circulation; limited specificity for glymphatic function	Tracer type; administration route; imaging timing; underlying CSF circulation abnormalities [64]
Meningeal lymphatic outflow imaging (emerging)	Human tracer-based approaches may involve IV or intrathecal GBCA; radio-tracer SPECT/CT evidence is primarily preclinical.	Mirrors tracer protocol; multi-timepoint	Dural enhancement and tracer-related lymphatic outflow patterns in human MRI studies [27, 65]; CSF radiotracer distribution and lymphatic outflow demonstrated in pre-clinical SPECT/CT studies [66]	Reproducibility remains insufficiently established because of nonstandardized protocols and subtle/nonspecific signals [27]	Emerging specialized research assessment; protocols and interpretation remain nonstandardized [27]	Imaging timing; tracer route/dose; anatomical ROI/segmentation; underlying CSF or disease characteristics [27, 65]

Abbreviations: *ALPS* diffusion tensor imaging along the perivascular space, *ASL* arterial spin labeling, *BBB* blood–brain barrier, *BOLD* blood oxygen level-dependent, *CEST* chemical exchange saturation transfer, *CSF* cerebrospinal fluid, *DGE* dynamic glucose-enhanced, *EEG* electroencephalography, *ePVS* enlarged perivascular spaces, *GBCA* gadolinium-based contrast agent, *IT* intrathecal, *IV* intravenous, *IVIM* intravoxel incoherent motion, *LP* lumbar puncture, *MRI* magnetic resonance imaging, *PVS* perivascular spaces, *ROI* region of interest, *SNR* signal-to-noise ratio, *TE* echo time, *VENO* velocity encoding

Reproducibility is described qualitatively in terms of reported methodological sources of variability rather than using categorical low/medium/high ratings, because standardized quantitative reproducibility estimates are not consistently available across modalities. Applicability refers to the current use of each modality for glymphatic-related assessment rather than the general clinical availability of the underlying imaging technique. Although structural MRI, diffusion MRI, conventional ASL, and other modalities may be routinely available clinically, their glymphatic-related or exchange-sensitive derived measures have not been validated for individual-level screening or diagnosis

Intravenous GBCA-enhanced MRI

Intravenous GBCA administration is clinically familiar and technically straightforward, but limited penetration across an intact blood–brain barrier produces relatively subtle CSF or parenchymal enhancement [6]. Sensitive T1-weighted imaging, quantitative T1 mapping, MP2RAGE, or MR

fingerprinting may therefore be combined with serial delayed imaging over several hours. This substantially increases scanner time and analytical burden. Reproducibility depends on standardized injection-to-scan timing, blood–brain barrier status, physiological conditions, sequence parameters, and image registration across time points [6].

Intrathecal gadolinium MRI (IT-GBCA)

Intrathecal GBCA MRI requires lumbar puncture followed by serial T1-weighted imaging, commonly over 24–48 h [6, 26]. Because GBCA administration by the intrathecal route is off-label, protocols require specialized clinical oversight and careful safety procedures. Measurements are sensitive to tracer dose, imaging timing, and individual CSF dynamics, while tracer distribution and washout may also vary with underlying disease state [6, 27]. Repeated imaging sessions make the approach resource-intensive and low throughput, while reproducibility depends on strict standardization of administration and acquisition procedures. These requirements currently restrict IT-GBCA primarily to specialized research settings.

Nuclear medicine techniques

Radionuclide cisternography

Radionuclide cisternography involves intrathecal administration of a radiotracer followed by serial gamma-camera or SPECT imaging to evaluate CSF circulation. Conventional approaches have used ^{111}In -DTPA and $^{99\text{m}}\text{Tc}$ -DTPA, with serial imaging performed over several hours depending on the tracer and protocol [64]. The method involves both lumbar puncture and ionizing radiation, and conventional radionuclide cisternography has limitations in spatial resolution [64]. Standardized tracer administration and serial imaging timing are important for consistent interpretation.

Meningeal lymphatic outflow imaging

Meningeal lymphatic imaging (emerging)

Human meningeal lymphatic imaging remains technically heterogeneous and generally relies on tracer-based protocols. Seminal human work by Absinta et al. demonstrated that dural lymphatic vessels can be visualized in vivo using high-resolution T2-FLAIR and T1-weighted black-blood MRI following intravenous gadobutrol administration [65]. These vessels were visualized predominantly alongside the dural venous sinuses, providing imaging evidence of meningeal lymphatic structures in humans [65]. Other human approaches include high-resolution T1-weighted MRI following intrathecal GBCA administration [27]. Preclinical SPECT/CT studies have additionally investigated CSF-administered radiotracer distribution and lymphatic outflow [66]. Practical limitations of current human tracer-based MRI approaches include invasiveness, repeated imaging, cost, limited accessibility, and nonstandardized acquisition and analysis methods [27]. Standardization of tracer

administration, acquisition timing, anatomical ROI definitions, and analysis methods will be necessary before broader reproducible application.

Practical implementation considerations

Across MRI-based approaches, reproducibility depends on modality-specific standardization of acquisition and analysis, together with appropriate control of physiological state, motion, and ROI/segmentation procedures [6, 25, 28, 29, 63]. Modality-specific characteristics including invasiveness, scan burden, quantitative outputs, reproducibility, current applicability, and major confounders are summarized in Table 3. The integration of these approaches into a tiered assessment strategy, together with minimum acquisition, output, metadata, and reporting recommendations, is presented separately in Sect. "Integrated framework for human glymphatic assessment and reporting checklist".

Integrated framework for human glymphatic assessment and reporting checklist

Assessment question first: define the physiological domain and measurement directness

Human glymphatic assessment should first identify the physiological domain of interest: transport (CSF/perivascular movement), exchange (CSF-ISF mixing), or clearance outcome (downstream retention/removal-related changes), because each modality interrogates different components or correlates of the glymphatic pathway [3, 7, 12, 13]. These domains are not mutually exclusive, and individual modalities may provide information across multiple domains depending on acquisition timing, analytical approach, and the physiological variable being interpreted. Accordingly, the proposed domain relevance represents an interpretive association with the predominant physiological process informed by a measure rather than a validated or exclusive classification. The physiological domain should be considered separately from measurement directness, which ranges from tracer-informed measures to physiology-sensitive neurofluid proxies, indirect imaging correlates, and downstream biological correlates. Assignment of a measure to a domain therefore does not imply that it directly measures the corresponding glymphatic process. This domain-based organization is particularly important given ongoing uncertainty regarding the relative contributions of convective and diffusive transport and the boundaries between glymphatic and other neurofluid clearance pathways; it is therefore intended to organize measurable processes without assuming a single definitive mechanistic model [3, 7, 12, 13]. This

domain-first framework is intended to reduce conceptual mismatch between measures (diffusion anisotropy, functional coupling, and tracer kinetics) and support interpretation across heterogeneous protocols.

Tiered assessment strategy with proposed operational minimums (acquisition, outputs, metadata)

A tiered framework organizes complementary assessment approaches according to feasibility, mechanistic depth, and measurement directness [3, 6, 7]. The operational minimums outlined below represent proposed recommendations derived from the reviewed literature and recurring methodological considerations rather than established consensus requirements. The tiers are intended as a research-oriented organizational framework and should not be interpreted as levels of clinical validation or readiness; prospective validation, normative reference data, and individual-level clinical thresholds will be required before clinical implementation. An overview of the proposed tiered framework, integrating physiological domain, measurement directness, and invasiveness, is presented in Fig. 2.

Tier 1 (research-oriented scalable assessment: structure+diffusion correlates) This tier can be incorporated into routine brain MRI with minimal additional burden [6, 15]. Proposed minimum acquisition includes T2-weighted imaging for primary ePVS assessment, with FLAIR used as a complementary sequence to distinguish PVS from surrounding white matter lesions, and DTI sufficient for DTI-ALPS calculation, both of which should be interpreted as indirect imaging correlates rather than direct measures of glymphatic transport, and neither should currently be considered a validated individual-level clinical screening measure. T1-weighted imaging can provide an anatomical reference and support segmentation when available [15, 29]. Proposed minimum outputs include regional ePVS burden (basal ganglia and/or centrum semiovale) using a declared visual or automated method, together with ALPS values reported using explicit ROI definitions and predefined left/right handling [14, 25, 29]. Proposed minimum metadata includes age, sex, vascular risk factors, time of day, recent sleep, caffeine and alcohol intake, neuroactive medications, and motion/QC criteria with exclusion rules [9, 25, 34, 35].

Tier 2 (physiology-informed mechanistic assessment: functional coupling and/or CSF flow) Proposed minimum acquisition adds resting-state fMRI for BOLD–CSF coupling and/or CSF flow imaging (phase-contrast MRI or 4D flow), which provide physiology-sensitive information on neurofluid coupling or macroscopic CSF motion rather than direct

measures of CSF–ISF exchange or solute clearance, with physiological monitoring recommended (respiration plus ECG/PPG and CO₂ when feasible) [18, 28, 63]. Proposed minimum outputs include the BOLD–CSF coupling definition (CSF ROI, lag/correlation method), preprocessing, motion-control strategy, and, for flow imaging, compartment location, VENC settings, segmentation, and quantitative flow metrics [28, 63]. Arousal status (eyes open/closed and drowsiness or sleep likelihood) should be reported because vigilance substantially influences coupling metrics [5, 19, 39]. Proposed minimum metadata extends Tier 1 by including respiratory characteristics, cardiac rhythm, blood pressure or pulsatility, and sleep-disorder status because these variables can influence CSF motion and neurofluid coupling [5, 18, 31, 32]. Same-day exercise and recent stimulant or sedative exposure should also be recorded as potential state-related covariates [35, 49], although evidence for their direct effects on BOLD–CSF coupling remains limited.

Tier 3 (tracer-informed validation/research assessment: tracer kinetics) Proposed minimum acquisition uses intrathecal GBCA (off-label) or IV GBCA with delayed imaging or T1 mapping and predefined serial time points, because tracer kinetics are sensitive to imaging timing and individual CSF dynamics [6, 27]. These approaches provide tracer-informed information on selected components of CSF distribution and clearance and should not be interpreted as comprehensive measures of the entire glymphatic pathway. Proposed minimum outputs include uptake and clearance kinetics (time to peak, decay/half-life or model parameters, and AUC), explicit ROIs, timestamps relative to injection, tracer information (agent, dose, route, and injection parameters), and safety outcomes [6, 27]. Proposed minimum metadata extends Tier 2 by including tracer agent, dose and administration route, imaging timing, relevant CSF or disease characteristics, BBB integrity when applicable, and renal function for GBCA protocols [6, 27].

Proposed minimum reporting recommendations and confounder control (core vs. recommended)

The following items represent proposed minimum reporting recommendations derived from the reviewed literature and recurring methodological considerations rather than established consensus requirements. Proposed core reporting items for all tiers include age, sex, diagnosis, vascular risk profile, neuroactive medications and recent changes, time of day, prior sleep, recent caffeine, alcohol and exercise, scanner field strength, key sequence parameters, motion metrics, ROI definitions, model-fitting parameters, and software versions [5, 9, 25, 30, 34, 35, 49, 53, 63]. These items are proposed to improve interpretability and reproducibility

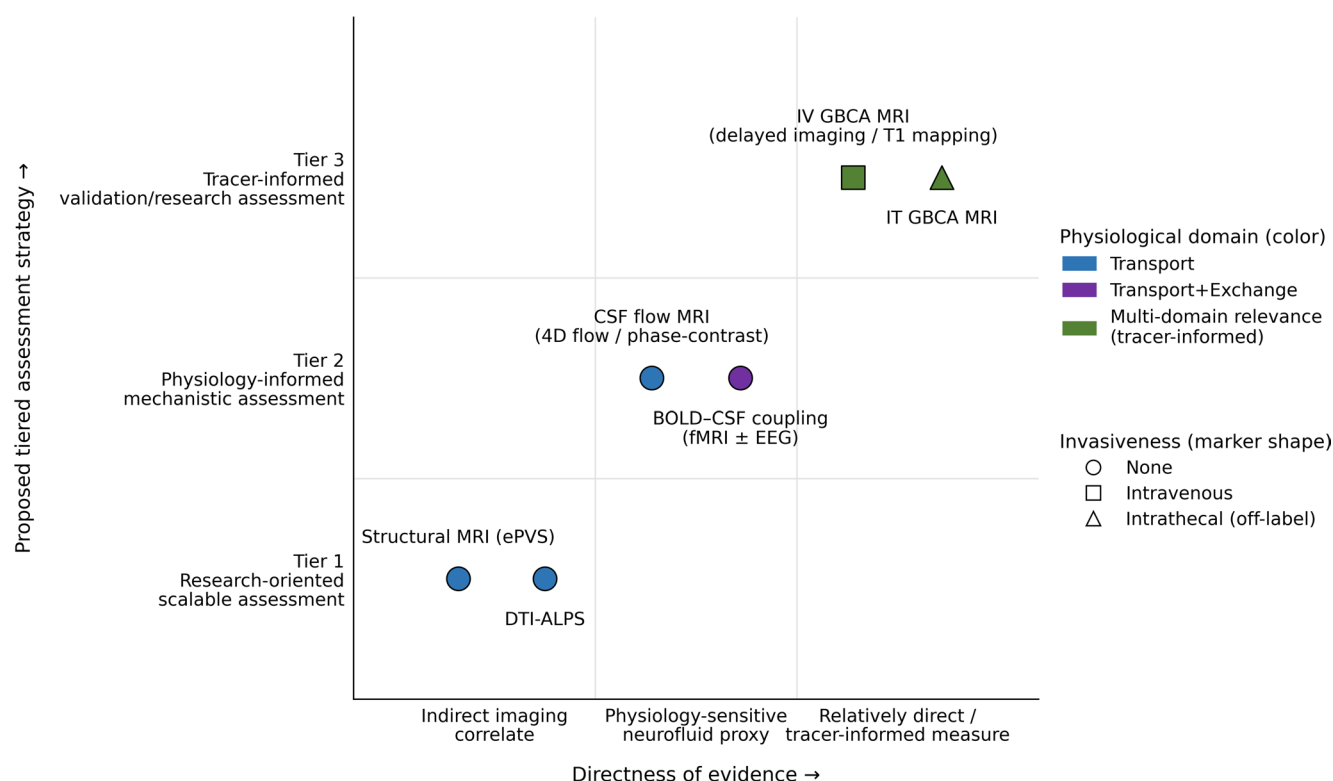


Fig. 2 Graphical summary of the proposed tiered framework for human glymphatic assessment, mapping representative modalities according to physiological domain (marker color), measurement directness (horizontal axis), invasiveness (marker shape), and proposed tiered assessment strategy (vertical axis). Tier 1 comprises indirect structural and diffusion correlates (structural MRI/ePVS and DTI-ALPS); Tier 2 comprises physiology-sensitive measures of functional coupling and/or CSF flow (BOLD–CSF coupling and phase-contrast MRI or 4D flow MRI); and Tier 3 comprises tracer-informed approaches (IV and intrathecal GBCA MRI). Physiological domain assignments represent proposed interpretive associations with the predominant process

informed by each modality rather than validated or exclusive classifications; tracer-informed approaches may provide information across multiple domains. The tiers represent the proposed research assessment framework rather than established clinical validation levels, and no modality is currently validated as a standalone individual-level glymphatic screening or diagnostic measure. Abbreviations: BOLD–CSF, blood oxygen level-dependent–cerebrospinal fluid coupling; DTI-ALPS, diffusion tensor imaging along the perivascular space; ePVS, enlarged perivascular spaces; GBCA, gadolinium-based contrast agent; IV, intravenous; MRI, magnetic resonance imaging

because vascular disease and aging may influence perivascular remodeling. Methodological sources of variability differ across modalities: DTI-ALPS is sensitive to acquisition and analysis conditions, including imaging plane, gradient directions, and ROI placement [25]; BOLD–CSF coupling estimates are influenced by arousal state, CSF ROI definition, preprocessing, motion, and the selected lag or correlation convention [19, 28, 39]; and CSF-flow MRI requires careful standardization of acquisition and quantitative flow analysis [63]. Within the proposed framework, Tier 2–3 studies should, when feasible, include respiration monitoring and CO₂/capnography because cardiorespiratory fluctuations influence CSF motion and neurofluid coupling [18, 31, 32].

Interpretation patterns for multimodal assessment

A structural-dominant pattern (high ePVS with limited Tier 2 abnormalities) may indicate chronic perivascular remodeling but is nonspecific and should be interpreted alongside

vascular risk factors and comorbidity burden [14, 29]. A diffusion/CSF-dynamics pattern (low ALPS and/or altered CSF flow) may be associated with altered diffusion characteristics or CSF dynamics but should not, by itself, be interpreted as evidence of impaired glymphatic transport [22, 25, 63]. A state/neurofluid-coupling pattern (reduced BOLD–CSF coupling under controlled physiology and motion) may indicate altered neurofluid coupling associated with arousal and sleep but does not directly quantify CSF–ISF exchange or solute clearance [5, 28]. Mixed abnormalities across tiers may strengthen evidence for altered neurofluid physiology when interpreted collectively, although they do not establish glymphatic dysfunction because the component measures differ in specificity and measurement directness [6, 7].

Translational applications and research utility

Within the proposed research framework, Tier 1 supports scalable cohort-level assessment, Tier 2 supports

physiology-informed mechanistic and state-dependent investigation, and Tier 3 supports tracer-informed validation and mechanistic investigation in specialized research settings. Standardized tier-specific acquisition, outcome reporting, and confounder control may improve comparability across studies and facilitate future development of multimodal composite scores and normative reference datasets.

Limitations, gaps, and future directions

Conceptual and mechanistic uncertainties

Important mechanistic uncertainties remain in the glymphatic model and should be considered when interpreting human measurements. The relative contributions of convective bulk flow and diffusion to parenchymal solute transport remain debated, as do the organization and relative roles of periaxonal and perivascular pathways [3, 7]. In addition, much of the foundational mechanistic evidence derives from animal tracer studies, and its direct translation to human neurofluid physiology remains uncertain [3, 7]. The proposed glymphatic pathway also operates alongside other neurofluid clearance mechanisms, including intramural periaxonal drainage and meningeal lymphatic outflow, and these pathways may overlap functionally rather than represent completely independent systems [7, 12, 13]. Accordingly, glymphatic-related measures in humans should be interpreted as reflecting selected components or correlates of a broader neurofluid clearance network rather than as definitive measurements of a single established pathway.

Why clinical adoption is still limited

Clinical translation remains limited by the lack of standardized multimodal assessment integrating transport, exchange, and clearance-related measures with different levels of measurement directness [3, 6, 7]. Intrathecal GBCA tracer MRI provides relatively direct information on selected components of CSF distribution and clearance but remains invasive, off-label, and impractical for routine use because of prolonged protocols [6, 27]. Intravenous GBCA approaches avoid lumbar puncture but provide subtler tracer-related signals and still require prolonged serial imaging [6]. Non-invasive methods, including DTI-ALPS, BOLD-CSF coupling, ePVS, and CSF-flow MRI, provide indirect or physiology-sensitive information and vary substantially in scalability and acquisition burden [5, 9, 15, 25, 28, 63]. Consequently, differences across studies are expected because modalities assess different physiological domains with different levels of specificity and directness rather than a single process [3, 7, 16].

Reproducibility barriers

Despite increasing methodological development, reproducibility remains a major barrier to human glymphatic assessment. Diffusion-, functional-, structural-, flow-, and tracer-based approaches remain sensitive to modality-specific acquisition and analytical choices, as detailed in Sect. "Measurement Strategies for Glymphatic Assessment" [5, 6, 15, 25, 28, 63]. In addition, inconsistent reporting of physiological state and major confounders, including sleep, respiration, CO₂, posture, vascular factors, and neuroactive substances, limits cross-study comparison [5, 9, 18, 30, 34, 35, 45]. Multicenter harmonization and standardized reporting are therefore priorities for future validation.

Missing normative values and clinically meaningful cutoffs

Many glymphatic-related measures, including DTI-ALPS, still lack broadly accepted age-stratified reference ranges and validated individual-level clinical thresholds, limiting individual interpretation and longitudinal monitoring [7, 9, 25]. Normative datasets should account for vascular risk and brain aging [30, 44, 56]. Clinically meaningful thresholds also require validation against cognition, disease severity, and CSF or PET pathology [14, 24, 58]. Without standardized reference data, integrated interpretation remains largely qualitative [6, 7, 29].

Future research priorities

Future work should establish consensus acquisition, analysis, and reporting standards for ALPS, BOLD-CSF coupling, CSF flow MRI, and tracer imaging. Studies should determine whether biomarkers respond consistently to controlled physiological modulators, including sleep, respiration, posture, and vascular interventions [5, 17, 32, 45]. Longitudinal and interventional studies are needed to establish clinical utility, while safer tracer techniques and standardized meningeal lymphatic imaging will improve translational feasibility [6, 27, 65, 66]. Multimodal integration should follow methodological harmonization and biological validation to preserve clinical interpretability [6, 7, 25, 63].

Conclusion

Research on human glymphatic-related physiology benefits from integrating complementary measures that differ in both the physiological domains they interrogate and their degree of measurement directness rather than treating any

single modality as a comprehensive measure [3, 7]. These measures should also be interpreted within the continuing mechanistic debate regarding glymphatic transport and its relationship to other neurofluid clearance pathways [3, 7, 12, 13]. DTI-ALPS, BOLD–CSF coupling, tracer-informed MRI, and ePVS provide complementary but non-equivalent information relevant to glymphatic-related physiology and should be interpreted according to both their physiological domain and evidence class [6, 9, 14, 28]. Because glymphatic-related measures are strongly influenced by physiological state, interpretation requires consideration of sleep, circadian timing, respiration, CO₂ dynamics, cardiovascular function, posture, medications, and aging [5, 18, 34, 35, 42, 45, 56]. This review proposes a research-oriented framework integrating measures with different levels of directness, physiological modulators, and measurement strategies to support study design, reproducibility, and interpretation across modalities [3, 6, 7]. Future clinical translation will require standardized acquisition and analysis, rigorous confounder reporting, normative reference datasets, and prospective longitudinal validation [6, 7, 9, 25, 27, 29, 63].

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Declarations

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